Sinehan Ezhilmuthu

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EDUCATION

University of California, Merced *Class of 2025*
Bachelor of Science in Computer Science, Dean's List Merced, CA

Relevant Coursework: Operating Systems, Distributed Software Systems, Computer Organization, Algorithm Design and Analysis, Data Structures, Database Systems, Object-Oriented Programming, Software Engineering, Intro to Artificial Intelligence, Full Stack Web Development, Digital Image Processing

San Joaquin Delta College *Class of 2022*
Associate of Arts and Science Stockton, CA

EXPERIENCE

AI Engineering Intern | *Python, Machine Learning, Data Pipelines, Log Analysis*
Omron Robotics & Safety Technologies *Aug 2025 – Dec 2025*

• Architected an AI-driven log analysis system to parse and summarize large-scale robotics system logs, enabling engineers to rapidly diagnose failures across high-volume test runs.
• Engineered a scalable log-processing pipeline that automatically detected and clustered recurring anomalies across 100K+ log records, significantly improving debugging and root-cause identification.
• Integrated machine learning models into the pipeline to surface patterns and operational insights from raw telemetry, reducing manual log analysis time by 20%.

Software Engineering Intern | *Python, Robotics Software, Automation, Agile Development*
Milano Technical Group *June 2025 – Aug 2025*

• Engineered automation tooling in Python to convert engineering design specifications into executable robot control code, reducing manual configuration steps and accelerating robotics deployment cycles by 30%.
• Redesigned rapid-prototyping workflows for additive manufacturing and robotics assembly testing, decreasing prototype iteration time by 25% and enabling faster validation of mechanical–software integrations.
• Partnered with cross-functional teams across software, hardware, and mechanical engineering to integrate robotics control systems and ship features within Agile sprint cycles, improving system deployment reliability across R&D builds.

Research Test Engineer | *Python, Docker, CI/CD, REST APIs, Pytest*
UC Merced School of Engineering *Aug 2023 – May 2025*

• Optimized backend services for the UniPoll distributed polling API using Python and Docker, reducing endpoint latency by 30% and improving reliability for real-time polling across 10K+ datasets.
• Architected CI/CD pipelines to automate build, testing, and deployment workflows for containerized services, cutting manual release overhead and increasing deployment frequency by 40%.
• Developed a comprehensive automated testing framework using Pytest with 100+ integration and unit tests, expanding regression coverage and reducing production defects by 25%

PROJECTS

Anchor | Tauri, React, TypeScript, Rust, SwiftUI

• Engineered a cross-platform desktop application using Tauri (Rust backend, React + TypeScript frontend) supporting Windows, macOS, and Linux, reducing runtime memory usage by ~70% compared to Electron-based alternatives.
• Designed a Rust-based state machine architecture to manage module lifecycles and inter-process communication, enabling deterministic state transitions and improving system stability across multi-module workflows.

Portfolio LLM (Retrieval-Augmented Generation System) | FastAPI, Vector Search, Sentence Transformers [GitHub]

• Created an end-to-end RAG backend using FastAPI and Qdrant that ingests documents, generates embeddings, and performs vector search to answer recruiter queries about technical experience.
• Improved semantic retrieval using Sentence Transformers and vector indexing, cutting query latency by 35% while returning context-grounded responses with source citations.

TECHNICAL SKILLS

AI/LLM Tools: Cursor IDE, Claude Code CLI, OpenAI Codex GUI
Languages: Python, Swift, C++, Rust, SQL, TypeScript, JavaScript, HTML/CSS, Java
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Spring Boot, SwiftUI, Tailwind CSS, Firebase
Developer Tools: Git, Docker, Postman, VS Code, Xcode, Cursor, Linux, GitHub Projects, Azure Boards
Cloud & Infrastructure: AWS, CI/CD, REST APIs, Containerization
Data & AI: Qdrant, Vector Embeddings, Ollama, Sentence Transformers, RAG Systems
Testing: Pytest, Jest, React Testing Library, XCTest